WIPRO NGA Program – 25SUB4508_LSP

Capstone Project Presentation – 13th Feb2026

Project Title Here - **Enterprise Concurrent Signal Broadcast Framework (ECSBF)**

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Introduction:

- A high-performance distributed communication framework for real-time signal propagation.
- Designed to enable secure and reliable communication across multiple networked nodes.
- Uses a centralized Engine to manage authentication, sessions, and broadcasting.
- Built with concurrency, fault tolerance, and encryption as core principles.
- Ensures synchronized, traceable, and secure enterprise communication.

PROBLEM STATEMENT:

- Distributed systems require secure and synchronized real-time communication.
- Managing multiple concurrent node connections is complex and error-prone.
- Unauthorized access can lead to signal injection and security breaches.
- Network failures may disrupt sessions and cause data inconsistency.
- Lack of centralized observability makes debugging and monitoring difficult

SOLUTION:

- Centralized ECSBF Engine manages all node communication securely.
- Global Identity Registry ensures only provisioned nodes can connect.
- Multi-threaded concurrency engine handles simultaneous signal requests.
- Encrypted signal broadcasting ensures confidentiality and integrity.
- Diagnostic logging provides observability and system transparency.

WORKFLOW:

- Endpoint Nodes connect to the ECSBF Engine via secure TCP/IP channels.
- Secure handshake verifies node identity before session establishment.
- Engine Control Process manages authentication and validation.
- Concurrency Engine processes signals from multiple nodes simultaneously.
- Signal Broadcaster distributes validated signals to all active nodes.
- Diagnostic Logger records system activities at multiple severity levels.

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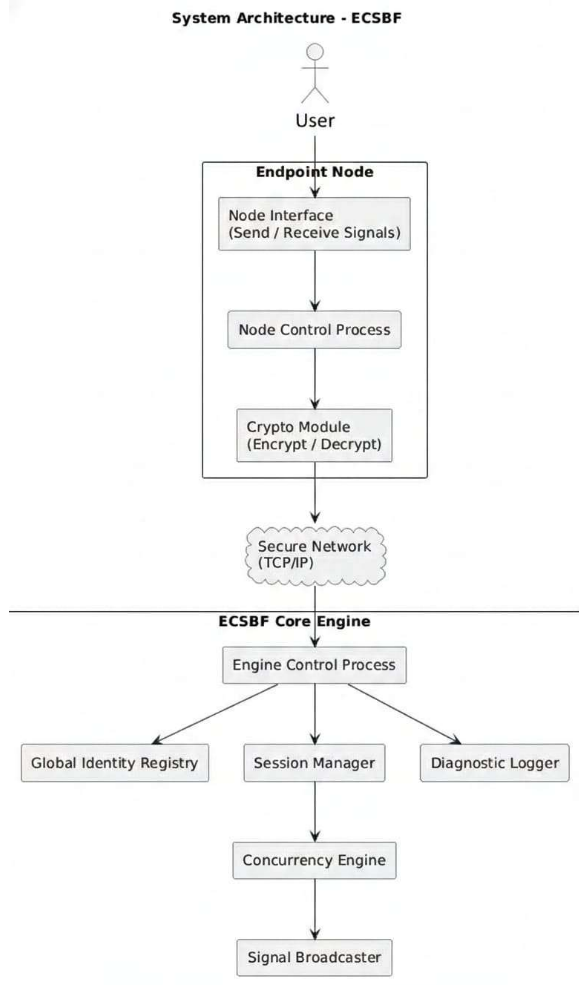
SYSTEM ARCHITECTURE DIAGRAM:



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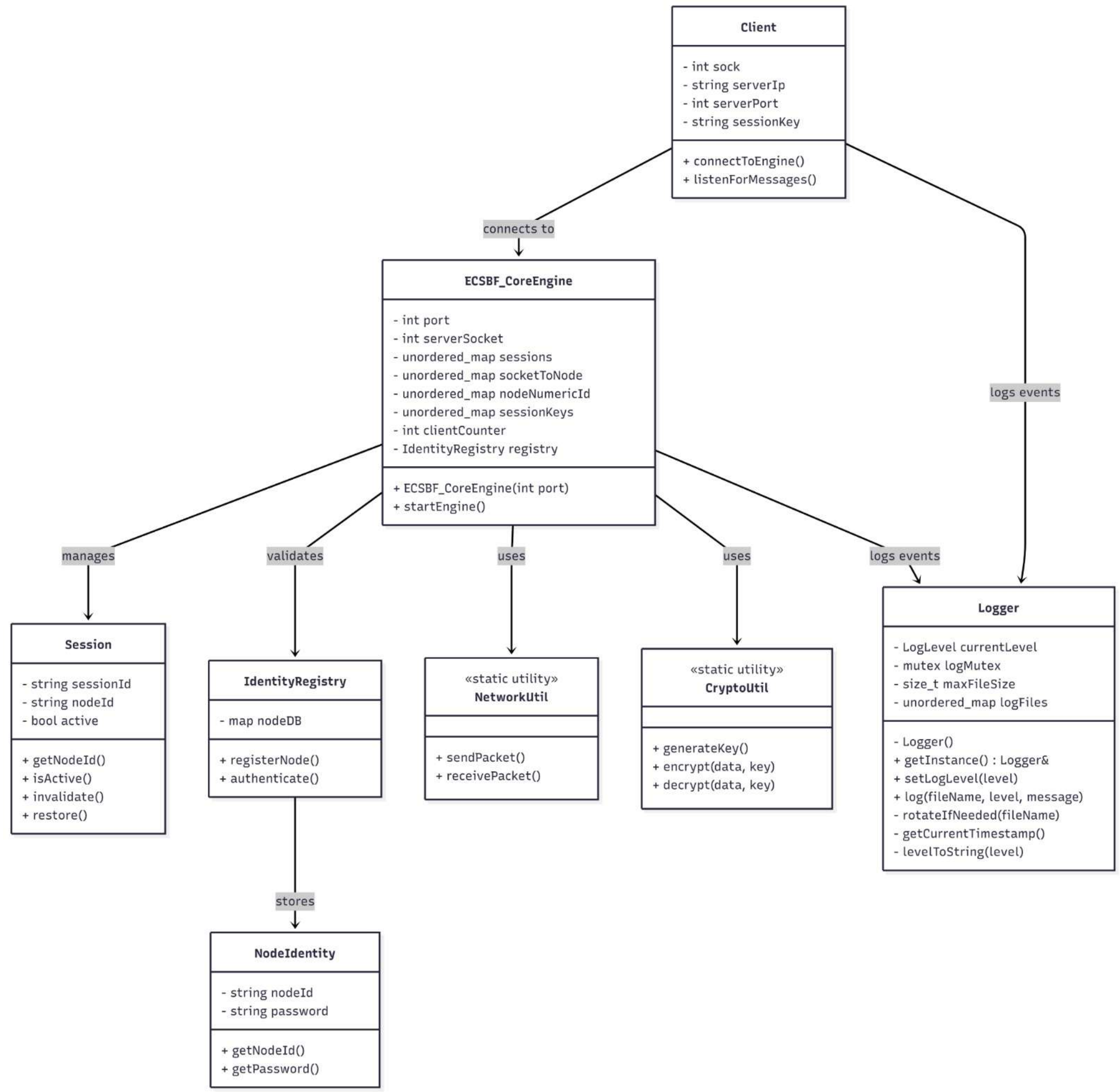
CLASS DIAGRAM:



PRACTICAL OUTCOMES – STEP 1 (Node Registration & Session Setup)

- Node initiates secure connection request to the Engine.
- Engine verifies identity from Global Identity Registry.
- Successful verification establishes a persistent session.
- Unauthorized nodes are rejected and logged securely.

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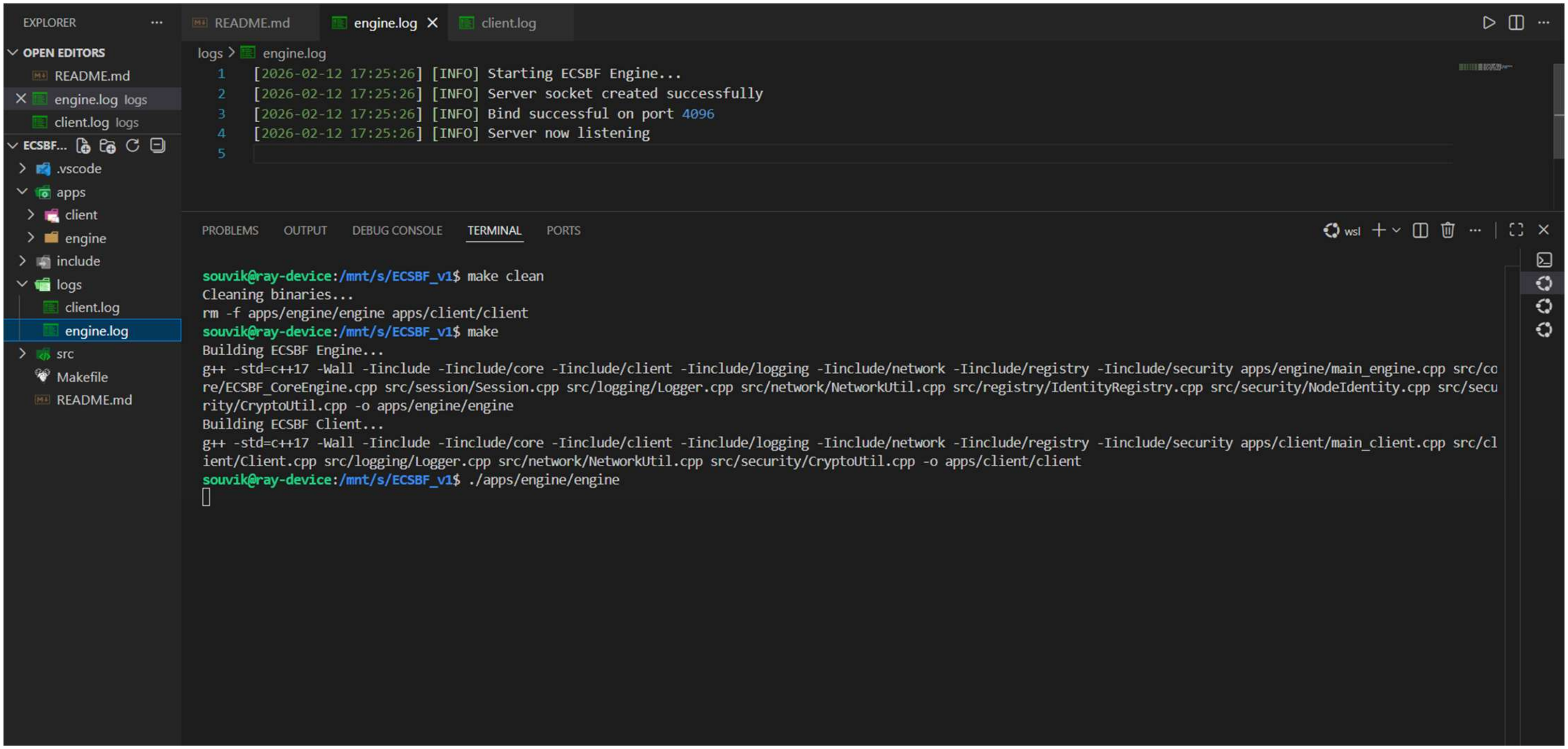

PRACTICAL OUTCOMES – STEP 2 (Signal Transmission & Broadcasting)

- Authenticated node sends encrypted signal payload to Engine.
- Engine validates active session before processing signal.
- Signal is tagged with source identifier for traceability.
- Broadcast mechanism sends signal to all active nodes concurrently.

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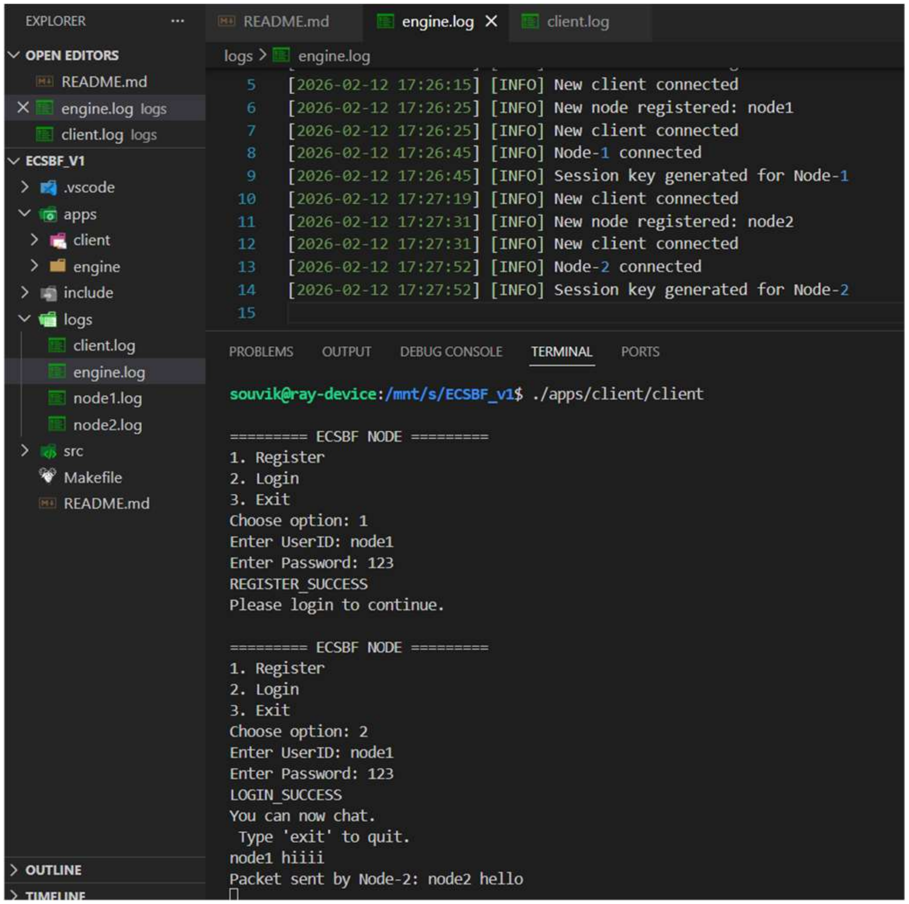

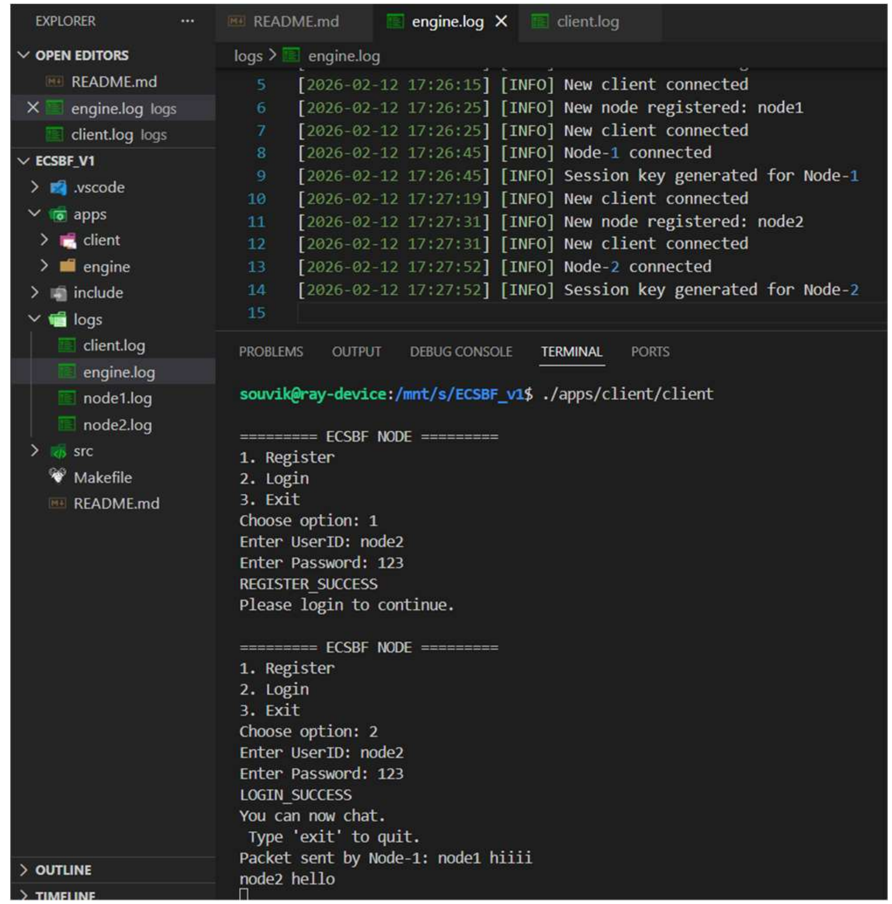

PRACTICAL OUTCOMES – STEP 3 (Concurrent Processing & Logging)

- Engine handles multiple node signals using multi-threading.
- Sessions remain stable under simultaneous traffic load.
- Diagnostic logs capture INFO, WARNING, DEBUG, and FATAL events.
- System maintains low latency and real-time propagation.

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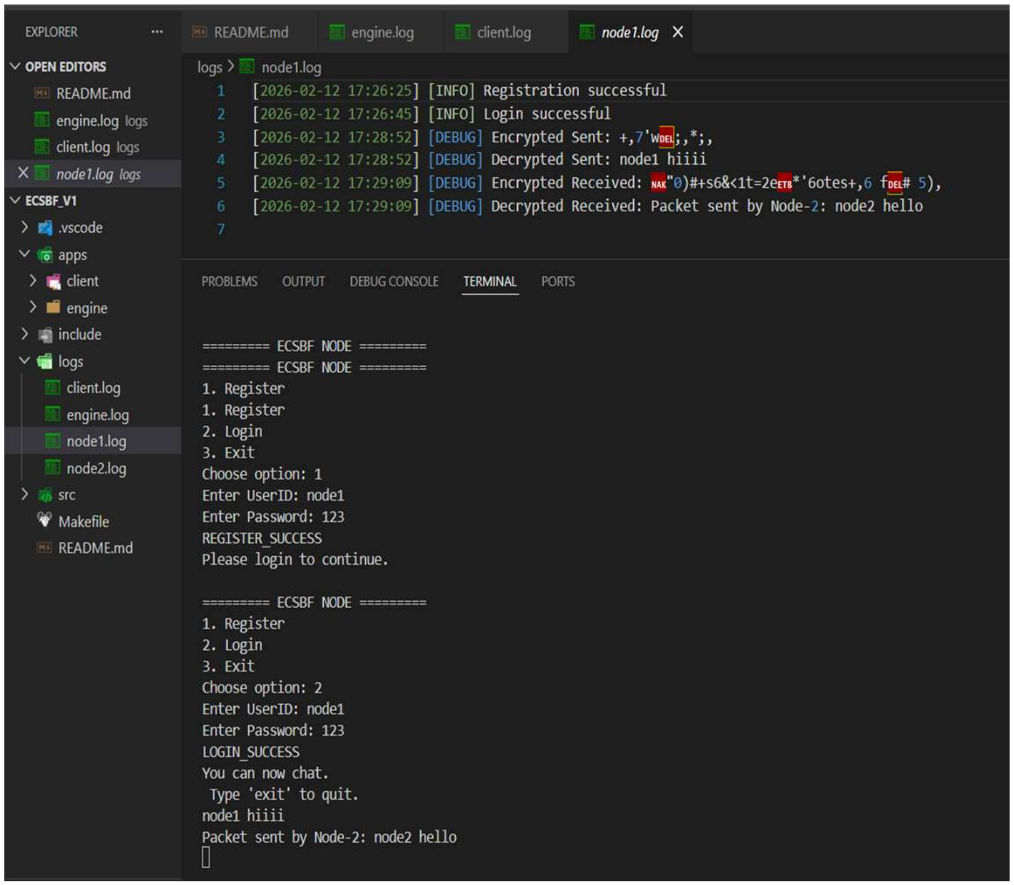

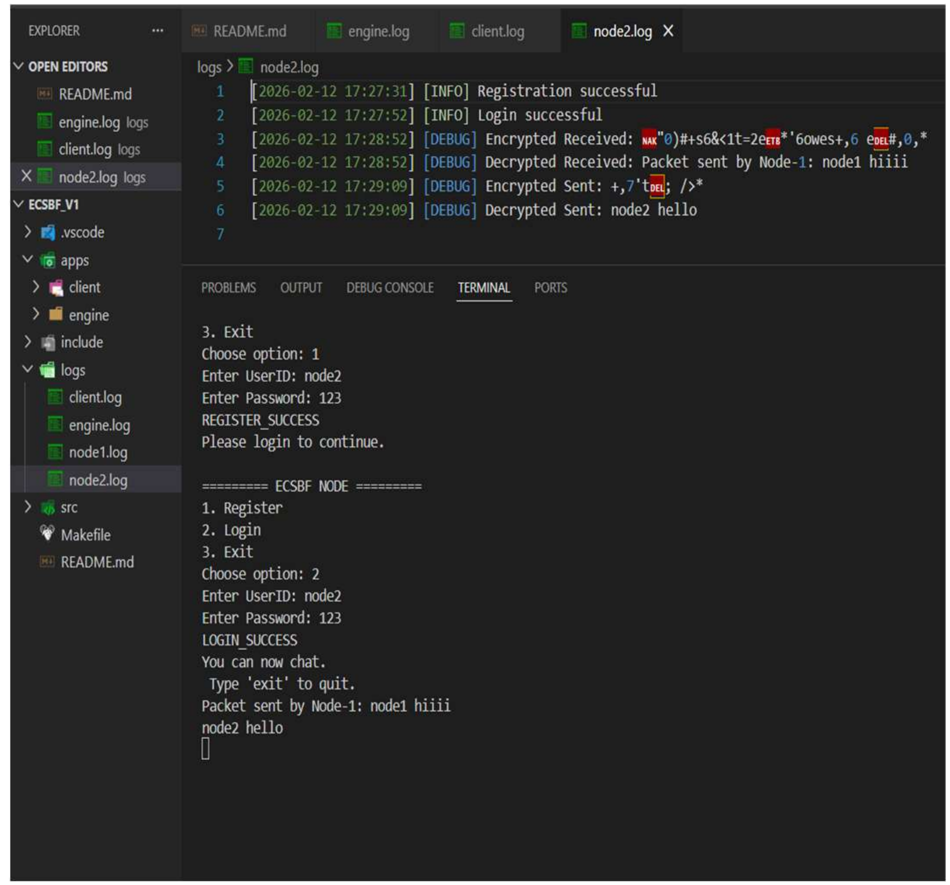

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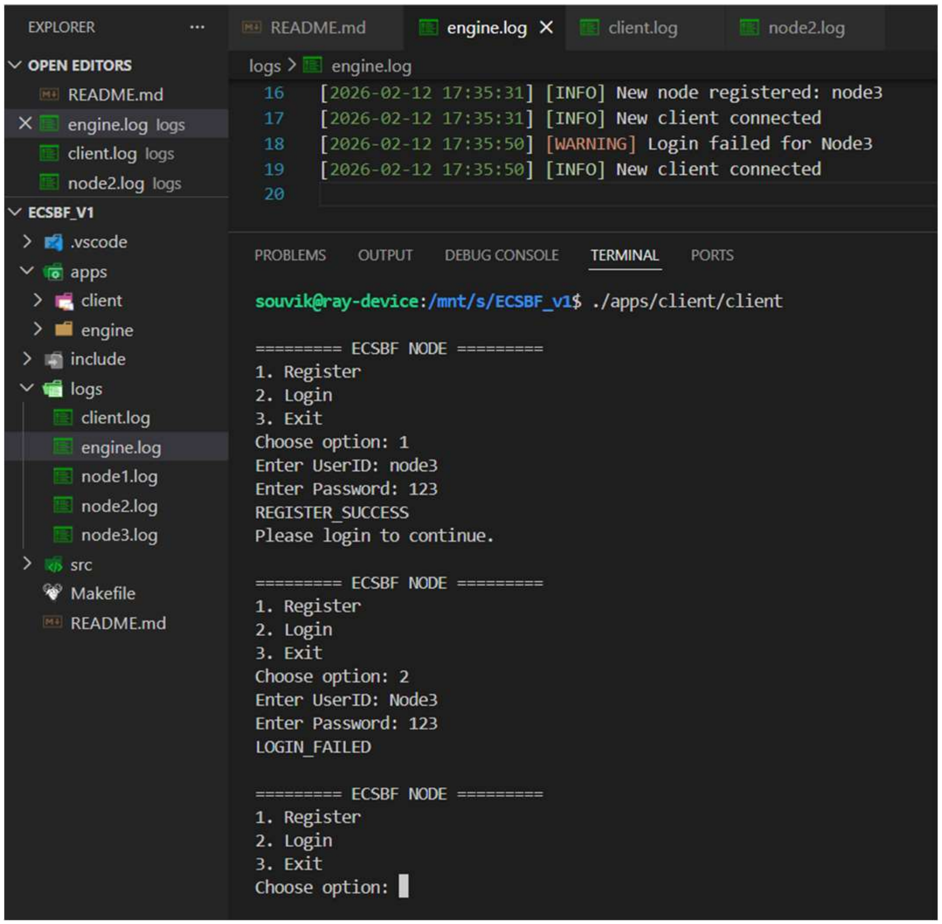

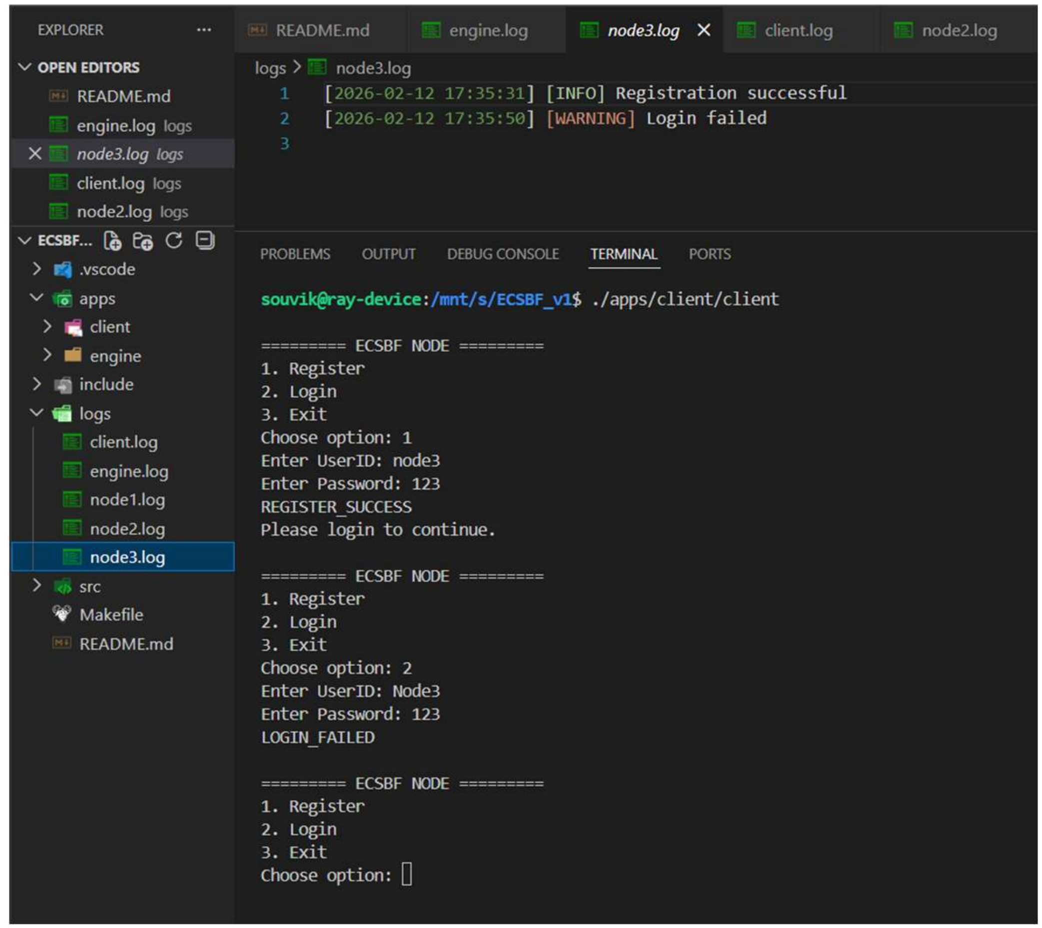

PRACTICAL OUTCOMES – STEP 4 (Session Termination)

- Engine detects node disconnection or timeout event.
- Session is formally terminated to prevent orphan processes.
- Logging mechanism records session closure event.

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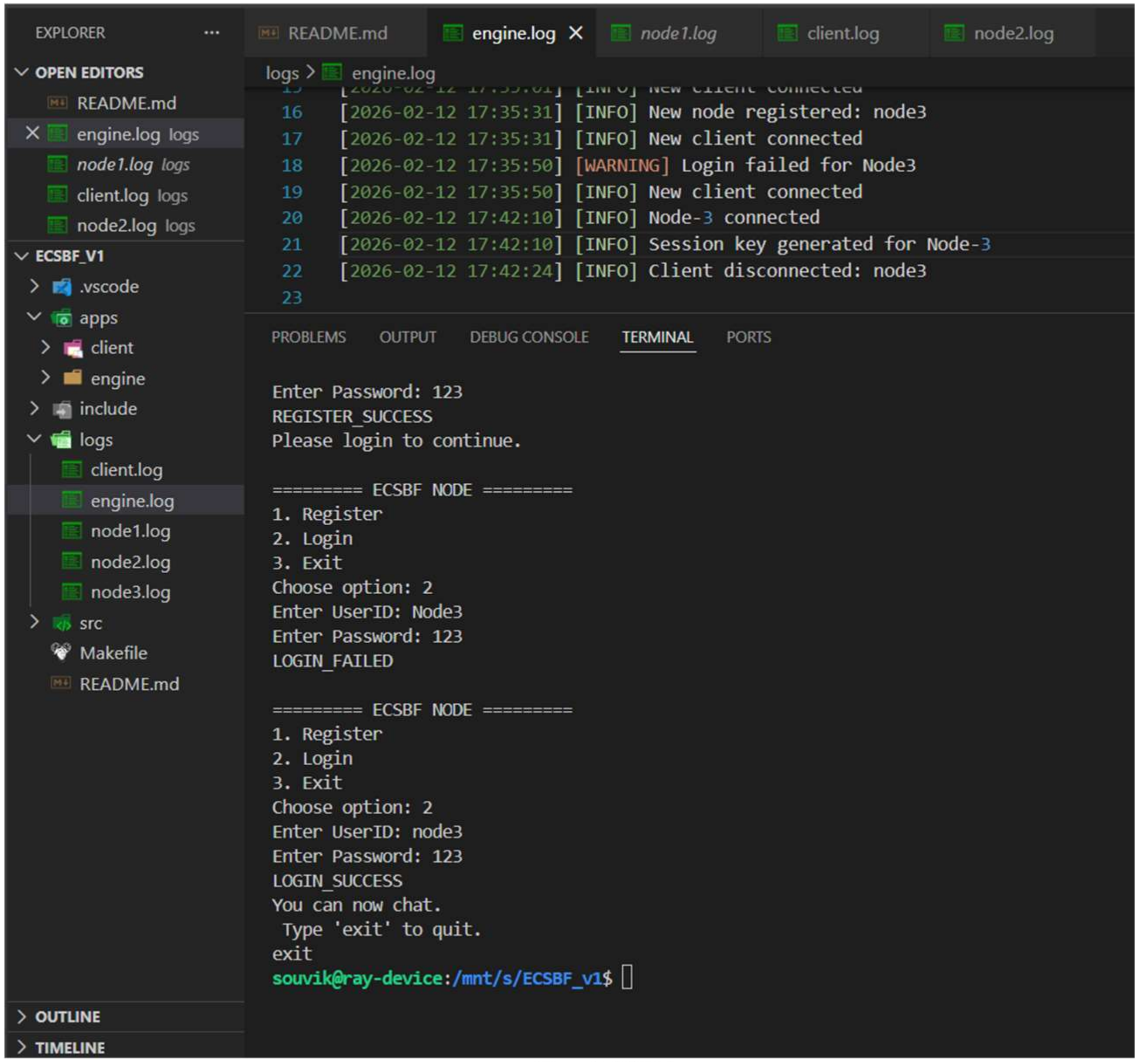

LIMITATIONS:

- Requires stable TCP/IP network connectivity for optimal performance.
- Centralized Engine may become bottleneck under extreme scale.
- Currently backend-focused with no dedicated graphical UI.
- Performance depends on underlying hardware and OS thread management.
- Encryption overhead may slightly increase processing latency.

CONCLUSION:

- ECSBF provides secure, scalable, and concurrent signal broadcasting.
- Strong identity verification prevents unauthorized system access.
- Multi-threaded architecture ensures high-performance communication.
- Persistent session management enhances reliability and resilience.
- Modular and extensible design supports future enterprise enhancements.

Thank You!

	SOUVIK ROY
SRS	Drafted Introduction, Purpose, and Project Scope sections defining overall system vision and objectives.
DESIGN	Designed Core Engine architecture including Concurrency Engine and Signal Broadcaster modules.
CODE	Implemented Engine Control Process, multi-threaded concurrency logic, and broadcast mechanism.
CUT	Performed Engine-side unit testing and validated concurrent signal processing stability.

KRITHIV DHARAN
Documented System Features including Secure Registration and Concurrent Broadcasting requirements.
Designed Global Identity Registry and Session Management workflow diagrams.
Implemented Node connection handling, session establishment, and session restoration logic.
Tested node connectivity, session recovery, and identity verification scenarios.

ALISHA PARVEEN
Prepared Non-Functional Requirements covering performance, security, scalability, and reliability aspects.
Designed Secure Handshake, Encryption/Decryption flow, and Payload Handling structure.
Implemented cryptographic modules for encryption, decryption, and secure payload transmission.
Tested encryption integrity, payload validation, and secure transmission accuracy.

RITTWIKA DATTA
Compiled all external interface details, project assumptions, glossary terms, formatting updates, and finalized the complete ECSBF document.
Designed the system architecture, logging structure, and complete message workflow for ECSBF.
Implemented logging with FATAL, INFO, WARNING, and DEBUG levels, and handled proper session termination when clients disconnect.
Integrated all ECSBF modules, tested full client–engine communication, fixed issues, and verified final system execution.